

AI ASSISTED CODING

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Lab 13: Code Refactoring – Improving Legacy Code with AI

Suggestions

Task Description -1 (Refactoring – Eliminating Repetitive Logic)

- Task: Use AI-assisted coding techniques to restructure a Python program that performs repeated calculations, transforming it into a reusable and maintainable design.

Instructions:

- Ask AI to locate repeated computations within the program.
- Encapsulate repeated logic into reusable functions.
- Verify that program output remains unchanged after refactoring.
- Add proper docstrings to all defined functions.

Sample Code:

```
length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
```

Expected Output:

A refactored Python script that uses a single reusable function to compute area and perimeter without repeating code.

```
Lab_13.2.py > ...
1  #TASK - 1
2  #restructure a Python program that performs repeated calculations, transforming it into a reusable and maintainable design.
3  length = 8
4  width = 4
5  print("Area:", length * width)
6  print("Perimeter:", 2 * (length + width))
7  length = 12
8  width = 6
9  print("Area:", length * width)
10 print("Perimeter:", 2 * (length + width))
11 # Refactored Code:
12 def calculate_area(length, width):
13     return length * width
14 def calculate_perimeter(length, width):
15     return 2 * (length + width)
16 length = 8
17 width = 4
18 print("Area:", calculate_area(length, width))
19 print("Perimeter:", calculate_perimeter(length, width))
20 length = 12
21 width = 6
22 print("Area:", calculate_area(length, width))
23 print("Perimeter:", calculate_perimeter(length, width))
24
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS Python De

```
PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC> & 'c:\Users\chand\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\chand\...
○ Perimeter: 24
Area: 72
Perimeter: 36
Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
```

ANALYSIS

- The program calculates the area and perimeter of a rectangle.
- Initially, the same calculations are repeated in the code.
- Functions `calculate_area()` and `calculate_perimeter()` are created to avoid repetition.
- This makes the code reusable, cleaner, and easier to maintain.

Task Description -2 (Refactoring – Modularizing Inline

Calculations)

- Task: Use AI to refactor a Python script by extracting repeated inline calculations into reusable and well-documented functions.

Instructions:

- Identify repeated calculation patterns in the code.
- Convert them into reusable functions.
- Ensure readability and proper documentation.

Sample Code:

```
amount = 1000
```

```
discount = amount * 0.10
```

```
final_amount = amount - discount
```

```
print("Final Amount:", final_amount)
```

```
amount = 2000
```

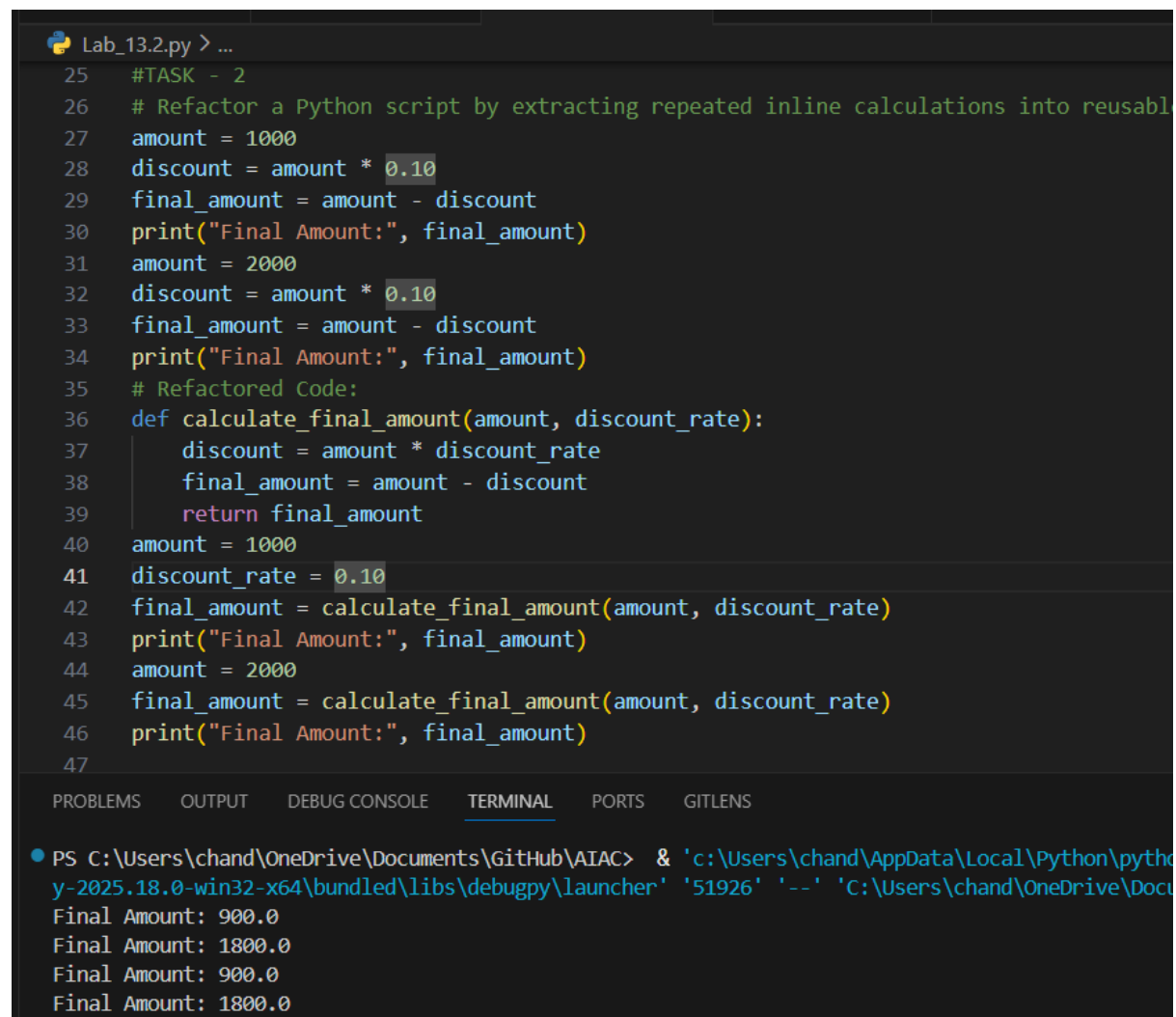
```
discount = amount * 0.10
```

```
final_amount = amount - discount
```

```
print("Final Amount:", final_amount)
```

Expected Output:

A modular Python program that uses a reusable function to compute final amounts for different inputs.



```
Lab_13.2.py > ...
25 #TASK - 2
26 # Refactor a Python script by extracting repeated inline calculations into reusable
27 amount = 1000
28 discount = amount * 0.10
29 final_amount = amount - discount
30 print("Final Amount:", final_amount)
31 amount = 2000
32 discount = amount * 0.10
33 final_amount = amount - discount
34 print("Final Amount:", final_amount)
35 # Refactored Code:
36 def calculate_final_amount(amount, discount_rate):
37     discount = amount * discount_rate
38     final_amount = amount - discount
39     return final_amount
40 amount = 1000
41 discount_rate = 0.10
42 final_amount = calculate_final_amount(amount, discount_rate)
43 print("Final Amount:", final_amount)
44 amount = 2000
45 final_amount = calculate_final_amount(amount, discount_rate)
46 print("Final Amount:", final_amount)
47
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC> & 'c:\Users\chand\AppData\Local\Python\python-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '51926' '--' 'C:\Users\chand\OneDrive\Documents\GitHub\AIAC\Lab_13.2.py'
Final Amount: 900.0
Final Amount: 1800.0
Final Amount: 900.0
Final Amount: 1800.0
```

ANALYSIS

-
- The program calculates the final amount after a discount.
 - Initially, the same discount calculation is repeated in the code.
 - A function `calculate_final_amount()` is created to avoid repetition.
 - This makes the code cleaner, reusable, and easier to maintain.
-

Task Description -3 (Refactoring – Improving Loop and Conditional Performance)

- Task: Apply AI-based suggestions to enhance the performance of Python code that relies on nested loops and repeated conditional checks.

Instructions:

- Use AI to recommend more efficient data structures or logic simplifications.
- Replace inefficient constructs while preserving program behavior.
- Compare execution efficiency before and after refactoring.

Sample Code:

```
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]

for name in check_list:
    exists = False

    for student in students:
        if name == student:
            exists = True

    if exists:
        print(name, "found")
    else:
        print(name, "not found")
```

Expected Output:

An optimized Python program that performs faster lookups using

efficient logic, along with execution performance comparison.

```
Lab_13.2.py > ...
47
48 #TASK - 3
49 #Apply AI-based suggestions to enhance the performance of Python code that relies on nested loops
50 '''students = ["Anil", "Bhavya", "Charan", "Divya"]
51 check_list = ["Charan", "Esha", "Bhavya"]
52 for name in check_list:
53     exists = False
54     for student in students:
55         if name == student:
56             exists = True
57         if exists:
58             print(name, "found")
59         else:
60             print(name, "not found")'''
61 # Refactored Code:
62 students = ["Anil", "Bhavya", "Charan", "Divya"]
63 check_list = ["Charan", "Esha", "Bhavya"]
64 for name in check_list:
65     if name in students:
66         print(name, "found")
67     else:
68         print(name, "not found")
69
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC> & 'c:\Users\chand\AppData\Local\Python\pythoncore-3.14-
y-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58672' '--' 'C:\Users\chand\OneDrive\Documents\GitHub
Charan found
Esha not found
Bhavya found
PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC>
```

ANALYSIS

- The program checks whether names in `check_list` exist in the students list.
- Initially, it uses nested loops, which is slower.
- The refactored code uses `if name in students` to check directly.
- This improves performance and makes the code simpler.

Task Description -4 (Refactoring – Replacing Hardcoded Values with Constants)

- Task: Utilize AI assistance to improve maintainability by replacing hardcoded numerical values with named constants.

Instructions:

- Detect hardcoded numeric values used directly in expressions.

- Define descriptive constants at the beginning of the program.
- Update calculations to use these constants without changing results.

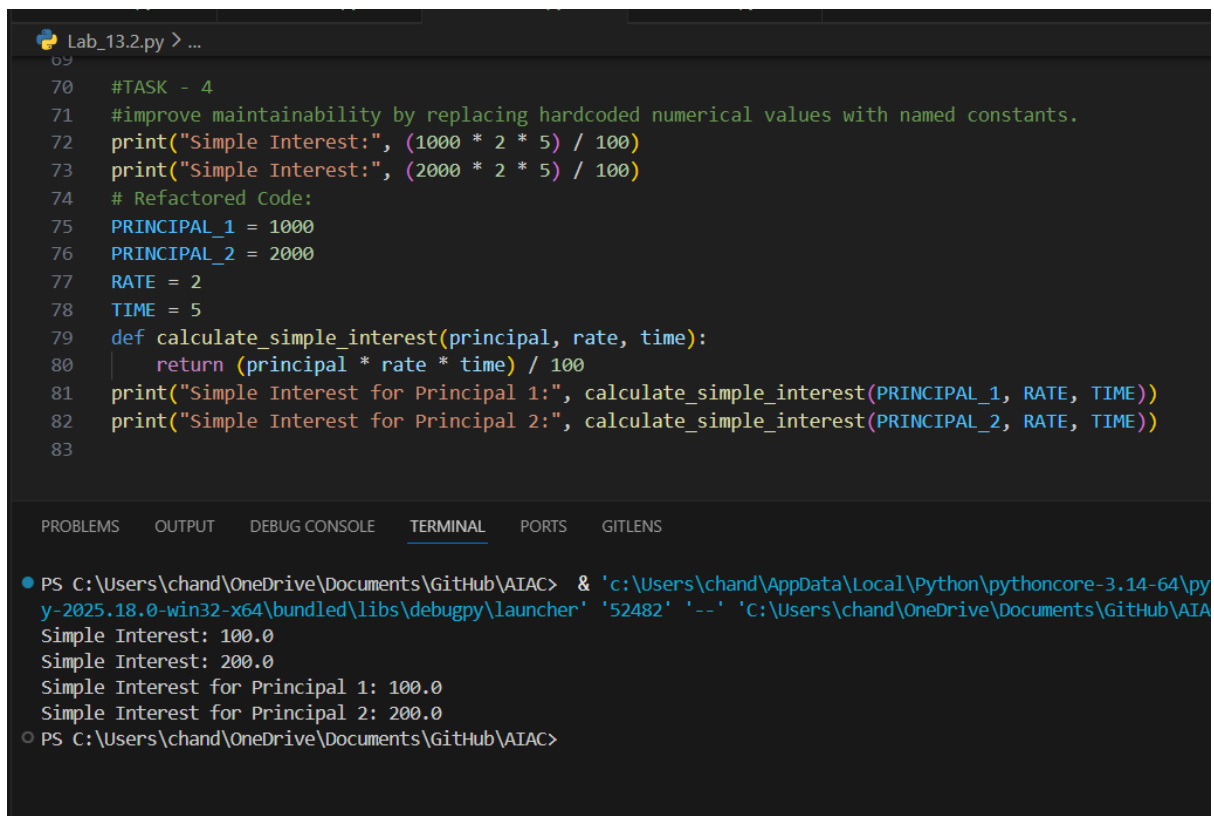
Sample Code:

```
print("Simple Interest:", (1000 * 2 * 5) / 100)
```

```
print("Simple Interest:", (2000 * 2 * 5) / 100)
```

Expected Output:

A revised Python program using constants such as RATE and TIME for clearer and maintainable calculations..



The screenshot shows a VS Code editor window with a file named 'Lab_13.2.py'. The code is as follows:

```
69
70 #TASK - 4
71 #improve maintainability by replacing hardcoded numerical values with named constants.
72 print("Simple Interest:", (1000 * 2 * 5) / 100)
73 print("Simple Interest:", (2000 * 2 * 5) / 100)
74 # Refactored Code:
75 PRINCIPAL_1 = 1000
76 PRINCIPAL_2 = 2000
77 RATE = 2
78 TIME = 5
79 def calculate_simple_interest(principal, rate, time):
80     return (principal * rate * time) / 100
81 print("Simple Interest for Principal 1:", calculate_simple_interest(PRINCIPAL_1, RATE, TIME))
82 print("Simple Interest for Principal 2:", calculate_simple_interest(PRINCIPAL_2, RATE, TIME))
83
```

The terminal output at the bottom shows the execution of the script:

```
PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC> & 'c:\Users\chand\AppData\Local\Python\pythoncore-3.14-64\python-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '52482' '--' 'C:\Users\chand\OneDrive\Documents\GitHub\AIAC\Lab_13.2.py'
Simple Interest: 100.0
Simple Interest: 200.0
Simple Interest for Principal 1: 100.0
Simple Interest for Principal 2: 200.0
PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC>
```

ANALYSIS

- The program calculates **Simple Interest**.
- Initially, it uses **hardcoded values** in the formula.
- The refactored code uses **named constants and a function** `calculate_simple_interest()`.
- This makes the code **clearer, reusable, and easier to maintain**.

Task Description -5 (Re factoring – Enhancing Variable Naming and Code Clarity)

- Task: Apply AI tools to improve code readability by renaming unclear variables and adding meaningful comments..

Instructions:

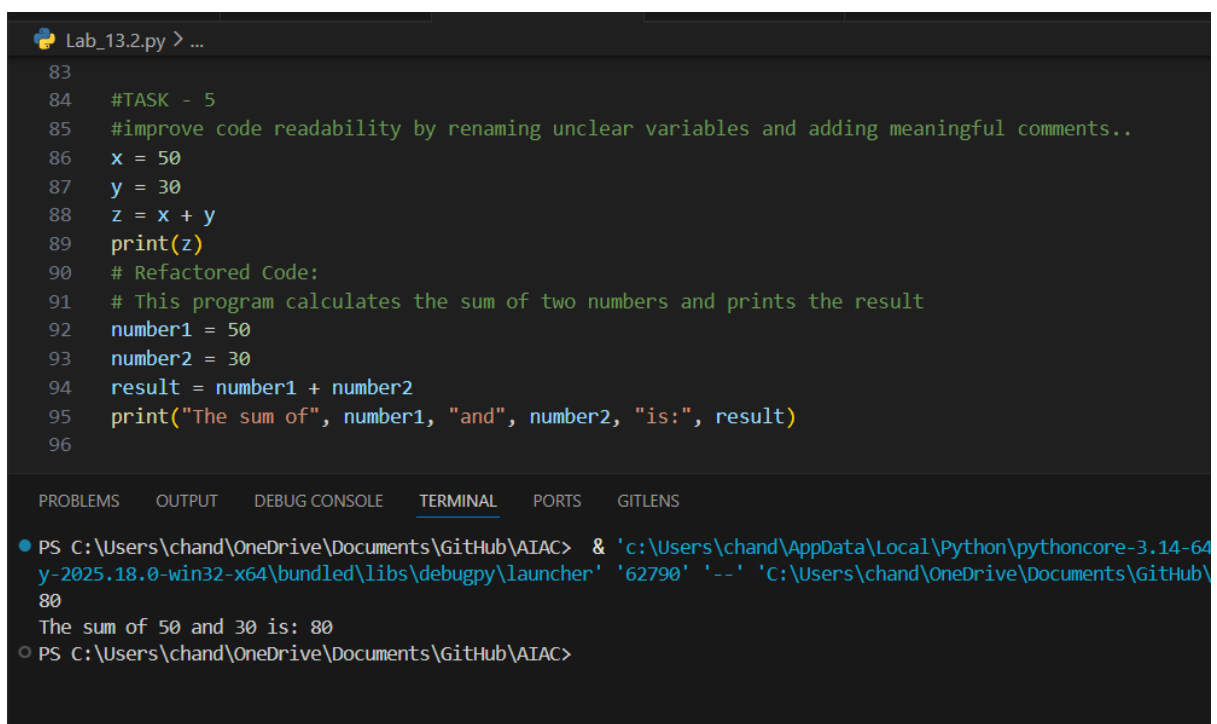
- Replace non-descriptive variable names with meaningful identifiers.
- Add inline comments where the logic may be unclear.
- Ensure program output remains the same.

Sample Code:

```
x = 50
y = 30
z = x + y
print(z)
```

Expected Output:

A Python program with clear variable names and explanatory comments that improve readability without altering functionality.



```
Lab_13.2.py > ...
83
84 #TASK - 5
85 #improve code readability by renaming unclear variables and adding meaningful comments..
86 x = 50
87 y = 30
88 z = x + y
89 print(z)
90 # Refactored Code:
91 # This program calculates the sum of two numbers and prints the result
92 number1 = 50
93 number2 = 30
94 result = number1 + number2
95 print("The sum of", number1, "and", number2, "is:", result)
96

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
● PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC> & 'c:\Users\chand\AppData\Local\Python\pythoncore-3.14-64
y-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '62790' '--' 'C:\Users\chand\OneDrive\Documents\GitHub\
80
The sum of 50 and 30 is: 80
○ PS C:\Users\chand\OneDrive\Documents\GitHub\AIAC>
```

ANALYSIS

- The program **adds two numbers and prints the result.**
 - Initially, **unclear variable names** (x, y, z) are used.
 - The refactored code uses **meaningful variable names** and comments.
 - This **improves readability and understanding of the code.**
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